

Capital Asset Pricing Model (CAPM): Definition, Formula, and Assumptions

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What Is the Capital Asset Pricing Model (CAPM)?

The capital asset pricing model (CAPM) describes the relationship between systematic risk, or the general perils of investing, and expected return for assets, particularly stocks. It is a finance model that establishes a linear relationship between the required return on an investment and risk.

CAPM is based on the relationship between an asset's beta, the risk-free rate (typically the Treasury bill rate), and the equity risk premium, or the expected return on the market minus the risk-free rate.

CAPM evolved as a way to measure this systematic risk. It is widely used throughout finance for pricing risky securities and generating expected returns for assets, given the risk of those assets and cost of capital.

Key Takeaways

- The capital asset pricing model, or CAPM, is a financial model that calculates the expected rate of return for an asset or investment.
- CAPM does this by using the expected return on both the market and a risk-free asset, and the asset's correlation or sensitivity to the market (beta).
- There are some limitations to the CAPM, such as making unrealistic assumptions and relying on a linear interpretation of risk vs. return.
- Despite its issues, the CAPM formula is still widely used because it is simple and allows for easy comparisons of investment alternatives.
- For instance, it is used in conjunction with modern portfolio theory (MPT) to understand portfolio risk and expected return.

Jessica Olah / Investopedia

Capital Asset Pricing Model (CAPM) Formula

The formula for calculating the expected return of an asset, given its risk, is as follows:

$ER_i = R_f + \beta_i(ER_m - R_f)$ where: ER_i = expected return of investment R_f = risk-free rate β_i = beta of the investment $(ER_m - R_f)$ = market risk premium

Investors expect to be compensated for risk and the time value of money. The risk-free rate in the CAPM formula accounts for the time value of money. The other components of the CAPM formula account for the investor taking on additional risk.

The goal of the CAPM formula is to evaluate whether a stock is fairly valued when its risk and the time value of money are compared with its expected return. In other words, by knowing the individual parts of the CAPM, it is possible to gauge whether the current price of a stock is consistent with its likely return.

The CAPM seeks to gauge whether a stock's current price is consistent with its likely return.

CAPM and Beta

Beta compares a security or portfolio's volatility or systematic risk to the market. If a stock is riskier than the market, it will have a beta greater than one. If a stock has a beta of less than one, the formula assumes it will reduce the risk of a portfolio.

A stock's beta is then multiplied by the market risk premium, which is the return expected from the market above the risk-free rate. The risk-free rate is then added to the product of the stock's beta and the market risk premium. The result should give an investor the required return or discount rate that they can use to find the value of an asset.

CAPM Example

Imagine an investor is contemplating a stock valued at \$100 per share today that pays a 3% annual dividend. Say this stock has a beta compared with the market of 1.3, which means it is more volatile than a broad market portfolio (i.e., the S&P 500 Index). Also, assume that the risk-free rate is 3% and this investor expects the market to rise in value by 8% per year.

The expected return of the stock based on the CAPM formula is 9.5%:

$$9.5\% = 3\% + 1.3 \times (8\% - 3\%)$$

The expected return of the CAPM formula is used to discount the expected dividends and capital appreciation of the stock over the expected holding period. If the discounted value of those future cash flows is equal to \$100, then the CAPM formula indicates the stock is fairly valued relative to risk.

Problems With the CAPM

Unrealistic Assumptions

Several assumptions behind the CAPM formula have been shown not to hold up in reality. Modern financial theory rests on two assumptions:

1. Securities markets are very competitive and efficient (that is, relevant information about the companies is quickly and universally distributed and absorbed).
2. These markets are dominated by rational, risk-averse investors, who seek to maximize satisfaction from returns on their investments.

As a result, it's not entirely clear whether CAPM works. The big sticking point is beta. When professors Eugene Fama and Kenneth French looked at share returns on the New York Stock Exchange (NYSE), the American Stock Exchange (AMEX), and Nasdaq, they found that differences in betas over a lengthy period did not explain the performance of different stocks. The linear relationship between beta and individual stock returns also breaks down over shorter periods of time. These findings seem to suggest that CAPM may be wrong.²

Including beta in the formula assumes that risk can be measured by a stock's price volatility. However, price movements in both directions are not equally risky. The look-back period to determine a stock's volatility is not standard because stock returns (and risk) are not normally distributed.

The CAPM also assumes that the risk-free rate will remain constant over the discounting period. Assume in the previous example that the interest rate on U.S. Treasury bonds rose to 5% or 6% during the 10-year holding period. An increase in the risk-free rate also increases the cost of the capital used in the investment and could make the stock look overvalued.

1960s

The period when the CAPM was conceived.²

Estimating the Risk Premium

The market portfolio used to find the market risk premium is only a theoretical value and is not an asset that can be purchased or invested in as an alternative to the stock. Most of the time, investors will use a major stock index, like the S&P 500, to substitute for the market, which is an imperfect comparison.

The most serious critique of the CAPM is the assumption that future cash flows can be estimated for the discounting process. If an investor could estimate the future return of a stock with a high level of accuracy, then the CAPM would not be necessary.

The CAPM and the Efficient Frontier

Using the CAPM to build a portfolio is supposed to help an investor manage their risk. If an investor were able to use the CAPM to perfectly optimize a portfolio's return relative to risk, it would exist on a curve called the efficient frontier, as shown in the following graph.

The graph shows how greater expected returns (y axis) require greater expected risk (x axis). Modern portfolio theory (MPT) suggests that starting with the risk-free rate, the expected return of a portfolio increases as the risk increases. Any portfolio that fits on the capital market line (CML) is better than any possible portfolio to the right of that line, but, at some point, a theoretical portfolio can be constructed on the CML with the best return for the amount of risk being taken.

The CML and the efficient frontier may be difficult to define, but they illustrate an important concept for investors: There is a tradeoff between increased return and increased risk. Because it isn't possible to perfectly build a portfolio that fits on the CML, it is more common for investors to take on too much risk as they seek additional return.

In the following chart, you can see two portfolios that have been constructed to fit along the efficient frontier. Portfolio A is expected to return 8% per year and has a 10% standard deviation or risk level. Portfolio B is expected to return 10% per year but has a 16% standard deviation. The risk of Portfolio B rose faster than its expected returns.

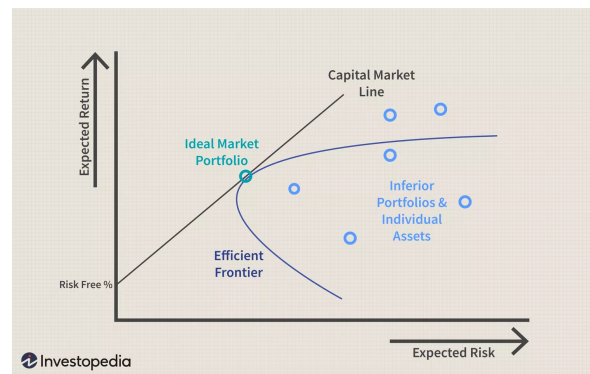


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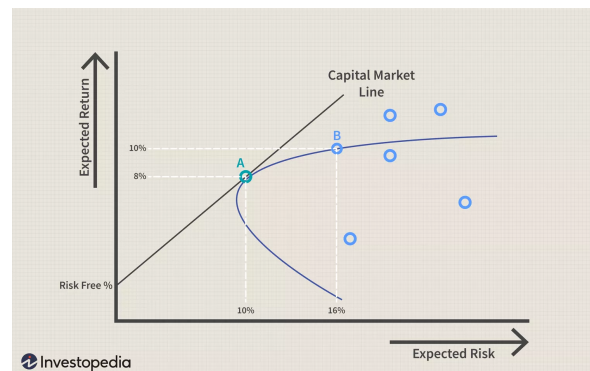


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CAPM and the Security Market Line (SML)

The efficient frontier assumes the same things as the CAPM and can only be calculated in theory. If a portfolio existed on the efficient frontier, it would provide maximal return for its level of risk. However, it is impossible to know whether a portfolio exists on the efficient frontier because future returns cannot be predicted.

This tradeoff between risk and return applies to the CAPM, and the efficient frontier graph can be rearranged to illustrate the tradeoff for individual assets. In the following chart, you can see that the CML is now called the security market line (SML). Instead of expected risk on the x axis, the stock's beta is used. As you can see in the illustration, as beta increases from 1 to 2, the expected return is also rising.

The CAPM and the SML make a connection between a stock's beta and its expected risk. Beta is found by statistical analysis of individual, daily share price returns compared with the market's daily returns over precisely the same period. A higher beta means more risk, but a portfolio of high-beta stocks could exist somewhere on the CML where the tradeoff is acceptable, if not the theoretical ideal.

The value of these two models is diminished by assumptions about beta and market participants that aren't true in the real markets. For example, beta does not account for the relative riskiness of a stock that is more volatile than the market with a high frequency of downside shocks compared with another stock with an equally high beta that does not experience the same kind of price movements to the downside.

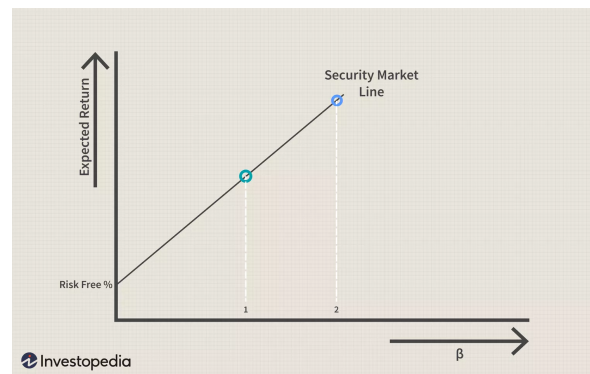


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Practical Value of the CAPM

Considering the critiques of the CAPM and the assumptions behind its use in portfolio construction, it might be difficult to see how it could be useful. However, using the CAPM as a tool to evaluate the reasonableness of future expectations or to conduct comparisons can still have some value.

Imagine an advisor who has proposed adding a stock to a portfolio with a \$100 share price. The advisor uses the CAPM to justify the price with a discount rate of 13%. The advisor's investment manager can take this information and compare it with the company's past performance and its peers to see if a 13% return is a reasonable expectation. Assume in this example that the peer group's performance over the last few years was a little better than 10%, while this stock had consistently underperformed, with 9% returns. The investment manager shouldn't take the advisor's recommendation without some justification for the increased expected return.

The CAPM can help generate ideas and reassess holdings but shouldn't be the only method used to value stocks.

An investor also can use the concepts from the CAPM and the efficient frontier to evaluate their portfolio or individual stock performance vs. the rest of the market. For example, assume that an investor's portfolio has returned 10% per year for the last three years with a standard deviation of returns (risk) of 10%. However, the market averages have returned 10% for the last three years with a risk of 8%.

The investor could use this observation to reevaluate how their portfolio is constructed and which holdings may not be on the SML. This could explain why the investor's portfolio is to the right of the CML. If the holdings that are either dragging on returns or have increased the portfolio's risk disproportionately can be identified, then the investor can make changes to improve returns.

Not surprisingly, the CAPM contributed to the rise in the use of indexing, or assembling a portfolio of shares to mimic a particular market or asset class, by risk-averse investors. This is largely due to the CAPM message that it is only possible to earn higher returns than those of the market as a whole by taking on higher risk (beta).³

Who Came Up With the CAPM?

The capital asset pricing model (CAPM) was developed in the early 1960s by financial economists William Sharpe, Jack Treynor, John Lintner, and Jan Mossin, who built their work on ideas put forth by Harry Markowitz in the 1950s.²

What Are Some of the Assumptions Built into the CAPM?

The following are assumptions made by the CAPM:

- All investors are risk-averse by nature.
- Investors have the same time period to evaluate information.
- There is unlimited capital to borrow at the risk-free rate of return.
- Investments can be divided into unlimited pieces and sizes.
- There are no taxes, inflation, or transaction costs.
- Risk and return are linearly related.

Many of these assumptions have been challenged as being unrealistic or plain wrong.⁴

What Are Some Alternatives to the CAPM?

Because of its criticisms, several alternative models to the capital asset pricing model (CAPM) have been developed to understand the relationship between risk and reward in investments. One of these is arbitrage pricing theory (APT), a model that looks at multiple factors, grouped into macroeconomic or company-specific factors.⁵ Another is the Fama-French 3-factor model, which expands on CAPM by adding company-size risk and value risk factors to the market risk factors.⁶

In 2015, Fama and French adapted their model to include five factors. Along with the original three factors, the new model adds the concept that companies reporting higher future earnings have higher returns in the stock market, a factor referred to as profitability. The fifth factor, referred to as “investment,” relates to the concept of internal investment and returns, suggesting that companies directing profit toward major growth projects are likely to experience losses in the stock market.⁷

What Is the International Capital Asset Pricing Model (ICAPM)?

The international capital asset pricing model (ICAPM) is a financial model that applies the traditional CAPM principle to international investments. It extends CAPM by considering the direct and indirect exposure to foreign currency in addition to time value and market risk included in the CAPM.

The Bottom Line

The CAPM uses the principles of modern portfolio theory to determine if a security is fairly valued. It relies on assumptions about investor behaviors, risk and return distributions, and market fundamentals that don't match reality.

However, the underlying concepts of CAPM and the associated efficient frontier can help investors understand the relationship between expected risk and reward as they strive to make better decisions about adding securities to a portfolio.